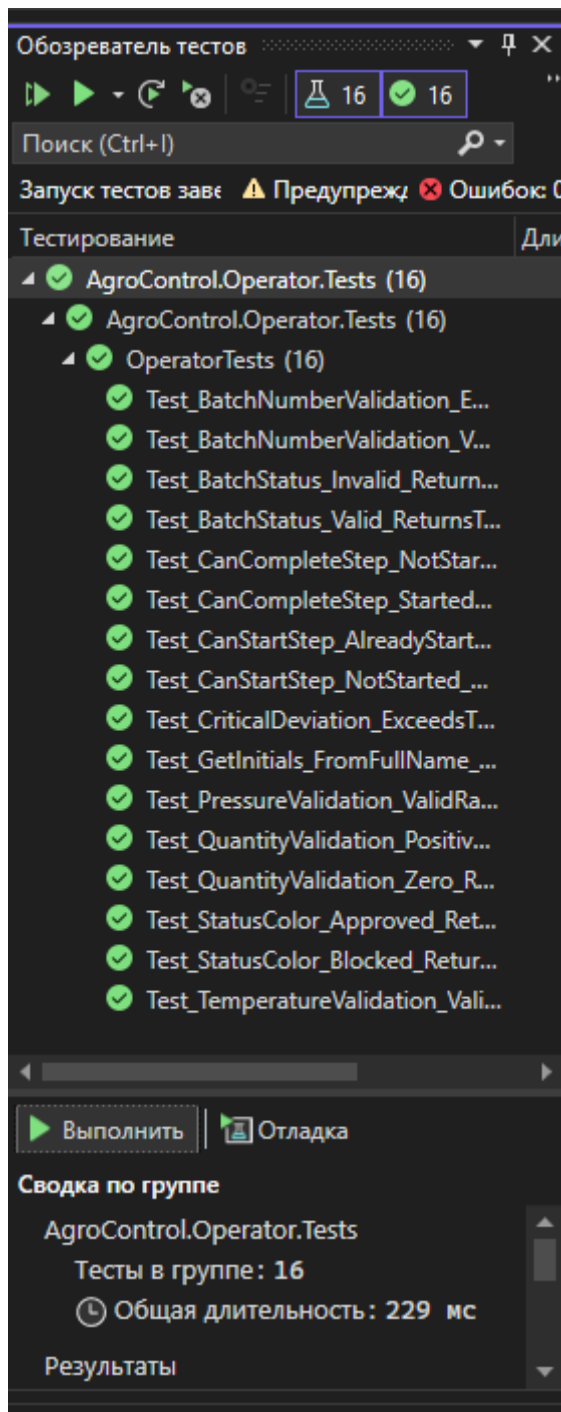




```
using Microsoft.VisualStudio.TestTools.UnitTesting;
```

```
using AgroControl.Operator;
```

```
namespace AgroControl.Operator.Tests
```

```
{
```

```
    [TestClass]
```

```
    public class OperatorTests
```

```
    {
```

[TestMethod]

```
public void Test_BatchNumberValidation_Valid_ReturnsTrue()
{
    bool isValid = TestHelpers.IsBatchNumberValid("B-2405-001");
    Assert.IsTrue(isValid);
}
```

[TestMethod]

```
public void Test_BatchNumberValidation_Empty_ReturnsFalse()
{
    bool isValid = TestHelpers.IsBatchNumberValid("");
    Assert.IsFalse(isValid);
}
```

[TestMethod]

```
public void Test_BatchStatus_Valid_ReturnsTrue()
{
    bool isValid = TestHelpers.IsValidBatchStatus("выполняется");
    Assert.IsTrue(isValid);
}
```

[TestMethod]

```
public void Test_BatchStatus_Invalid_ReturnsFalse()
{
    bool isValid = TestHelpers.IsValidBatchStatus("неизвестный");
    Assert.IsFalse(isValid);
}
```

[TestMethod]

```
public void Test_CanStartStep_NotStarted_ReturnsTrue()
{
    bool canStart = TestHelpers.CanStartStep("не начат");
    Assert.IsTrue(canStart);
}
```

[TestMethod]

```
public void Test_CanStartStep_AlreadyStarted_ReturnsFalse()
{
    bool canStart = TestHelpers.CanStartStep("выполняется");
    Assert.IsFalse(canStart);
}
```

[TestMethod]

```
public void Test_CanCompleteStep_Started_ReturnsTrue()
{
    bool canComplete = TestHelpers.CanCompleteStep("выполняется");
    Assert.IsTrue(canComplete);
}
```

[TestMethod]

```
public void Test_CanCompleteStep_NotStarted_ReturnsFalse()
{
    bool canComplete = TestHelpers.CanCompleteStep("не начат");
    Assert.IsFalse(canComplete);
}
```

[TestMethod]

```
public void Test_TemperatureValidation_ValidRange_ReturnsTrue()
```

```
{  
    bool valid = TestHelpers.IsTemperatureValid(85.5m);  
    Assert.IsTrue(valid);  
}
```

```
[TestMethod]  
public void Test_PressureValidation_ValidRange_ReturnsTrue()  
{  
    bool valid = TestHelpers.IsPressureValid(2.5m);  
    Assert.IsTrue(valid);  
}
```

```
[TestMethod]  
public void Test_QuantityValidation_Positive_ReturnsTrue()  
{  
    bool valid = TestHelpers.IsQuantityValid(500m);  
    Assert.IsTrue(valid);  
}
```

```
[TestMethod]  
public void Test_QuantityValidation_Zero_ReturnsFalse()  
{  
    bool valid = TestHelpers.IsQuantityValid(0);  
    Assert.IsFalse(valid);  
}
```

```
[TestMethod]  
public void Test_GetInitials_FromFullName_ReturnsCorrect()  
{
```

```

        string initials = TestHelpers.GetInitials("Зайцев Петр Николаевич");

        Assert.AreEqual("ЗП", initials);
    }

    [TestMethod]
    public void Test_CriticalDeviation_ExceedsThreshold_ReturnsTrue()
    {
        bool isCritical = TestHelpers.IsCriticalDeviation(100, 80, null, null, null, null);

        Assert.IsTrue(isCritical);
    }

    [TestMethod]
    public void Test_StatusColor_Approved_ReturnsGreen()
    {
        string color = TestHelpers.GetStatusColor("одобрена");

        Assert.AreEqual("Green", color);
    }

    [TestMethod]
    public void Test_StatusColor_Blocked_ReturnsDarkRed()
    {
        string color = TestHelpers.GetStatusColor("заблокирована");

        Assert.AreEqual("DarkRed", color);
    }
}

```